+++ layout = "page" title = "F-Algebras" date = 2019-02-09 [extra] banner="falgebras" +++

Let F be an endofunctor on the category Set. We define an **F-algebra** to be a pair (A, f) where $f\colon \mathbf{F}A \to A$.

Example: Group

A **group** is an $\mathbf{F_{Grp}}$-algebra where $\mathbf{F_{Grp}}A = 1 + A + A \times A$.

To see how to interpret the standard definition of a group, $(A, e, ^{-1}, \circ)$, in this language, observe that an element of the coproduct $\mathbf{F_{Grp}}A$ has one of three forms,

- $\mathrm{in}_0 1 : 1$ denotes the identity element of the group;
- $\mathrm{in}_1 x : A$ is an arbitrary element of the group's universe;
- $\mathrm{in}_2(x, y) : A \times A$ is an arbitrary pair of elements of the group's universe.

We define the group operations $f\colon \mathbf{F}A \to A$ by pattern matching as follows:

- $f\ \mathrm{in}_0 1 = e$ (the identity element of the group)
- $f\ \mathrm{in}_1 x = x^{-1}$ (the group's inverse operation)
- $f\ \mathrm{in}_2(x, y) = x \circ y$ (the group's binary operation)

Let (A, f) and (B, g) be two groups (i.e., $\mathbf{F_{Grp}}$-algebras).

A **homomorphism** from from (A, f) to (B, g), denoted by $h\colon (A, f) \to (B, g)$, is a function from A to B that satisfies the identity

$$h \circ f = g \circ \mathbf{F_{Grp}}h$$

To make sense of this, we must know how the functor $\mathbf{F_{Grp}}$ acts on arrows (i.e., homomorphisms, like h). It does so as follows:

- $(\mathrm{F}_{\mathbf{Grp}}\, h)(\mathrm{in}_0 1) = h(e)$;
- $(\mathrm{F}_{\mathbf{Grp}}\, h)(\mathrm{in}_1 x) = (h(x))^{-1}$;
- $(\mathrm{F}_{\mathbf{Grp}}\, h)(\mathrm{in}_2(x, y)) = h(x) \circ h(y)$.

Equivalently,

- $h \circ f(\mathrm{in}_0 1) = h(e)$ and $g \circ \mathbf{F_{Grp}}h(\mathrm{in}_0 1) = g(h(e))$;
- $h \circ f(\mathrm{in}_1 x) = h(x^{-1})$ and $g \circ \mathbf{F_{Grp}}h(\mathrm{in}_1 x) = g(\mathrm{in}_1 h(x)) = (h(x))^{-1}$;
- $h \circ f(\mathrm{in}_2(x, y)) = h(x \circ y)$ and $g \circ \mathbf{F_{Grp}}h(\mathrm{in}_2(x, y)) = g(\mathrm{in}_2(h(x), h(y))) = h(x) \circ h(y)$.

So, in this case, the indentity $h \circ f = g \circ \mathbf{F_{Grp}}h$ reduces to

- $h(e^A) = g(h(e))$;
- $h(x^{-1_A}) = (h(x))^{-1_B}$;
- $h(x \circ^A y) = h(x) \circ^B h(y)$,

which are precisely the conditions we would normally verify when checking that h is a group homomorphism.